Avinash Agarwal

Mathematics | Engineering | Artificial Intelligence | Distributed Systems

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The United States of America

Summary

Curiosity-driven engineering veteran with 8 years of industry-leading experience at Meta and Microsoft; specializing in full-stack development, distributed cloud infrastructure, and high-throughput data-pipelines. Proven track record of designing and managing high-volume distributed systems processing billions of daily notifications. Passionate about leveraging Artificial Intelligence and Distributed Systems to solve complex planet-scale engineering challenges.

Experience

Meta — *August 2025 - Present*
Software Engineer — Menlo Park, California, USA

- **Architecting and scaling Meta's global Credentials Infrastructure for Authentication Methods**, enhancing the security, reliability, and performance of identity systems processing **billions of authentication requests** across the entire Meta ecosystem.
- **Driving end-to-end product testing within Meta Superintelligence Labs**, ensuring stability, scalability, and peak performance for the Meta AI app.
- **Secured runners-up position in an internal Hackathon**, demonstrating high engineering velocity by rapidly diagnosing and resolving a high volume of bugs and edge cases across the codebase.
- **Improved on-call efficiency by integrating an automated on-call agent**, streamlining incident response and reducing manual triage for the engineering team.

Azure Core, Microsoft — *Feb 2022 - July 2025*
Software Engineer — Redmond, Washington, USA

Core Engineering | Azure Resource Graph

- **Significantly reduced partition skewness** in data-ingestion pipelines, achieving **over 30%** improvement in compute, memory and network efficiency.
- **Advocated for increasing hash cardinality across the storage stack**, significantly improving distribution granularity and partition skewness.
- **Engineered data ingestion components** hosted on the **Azure Synapse** platform to support dynamic time-window loading for security datasets.
- **Optimized complex SQL queries** executed on **Azure Synapse** to efficiently load and merge multiple continuous data streams.
- **Led migration of a streaming data pipeline (1B+ daily notifications)** to its dedicated scale unit — Involved sophisticated design, careful cross-team planning and execution.
- **Profiled and optimized high-contention services** eliminating bottlenecks and achieving **nearly 10%** improvement in compute, memory and network efficiency.
- **Improved on-call efficiency** with multiple improvements to internal monitors, documentations, etc.

Engineering Lead | Azure Resource Change Analysis

- **Delivered General Availability for the Change Analysis back-end** by deploying the entire service to US and China Government clouds, ahead of an important Gartner cloud review.
- **Led the architectural evolution, including metadata enhancements and optimizations** to the Change Analysis payload with extensive collaboration across teams and disciplines.
- **Delivered General Availability for the Change Actor** functionality, enabling customers to audit **who** initiated resource changes and **which client** (Portal, CLI, SDK) was used to make these changes.
- **Delivered General Availability for the Change Analysis front-end** (live on the Azure portal) by improving performance Lighthouse scores, Accessibility fixes, CI/CD, test automation setup etc.

Azure Global, Microsoft
Software Engineer

June 2018 - Jan 2022
Hyderabad, Telangana, India

Engineering Lead | Scaling for GRAB (Strategic Partnership)

- Collaborated with a specialist engineering partner to perform intensive load testing and performance profiling, ensuring platform stability for massive-scale deployment across Southeast Asia.
- Optimized core services to achieve a **20x throughput improvement** by studying and refining SQL query execution plans, eliminating redundant network calls, and implementing strategic caching layers.
- Architected APIs to execute the **migration of 160M+ customer records from AWS to Azure**, designing robust APIs to ensure zero-downtime and data integrity during the transition.
- Spearheaded technical design reviews and PM alignment, advocating for extensible solutions to avoid technical debt from short-term fixes.

Engineering Lead | UNICEF Learning Passport (powered by Microsoft Community Training)

- **Offline Innovation (Hackathon Lead):** Spearheaded a hackathon project to host the entire cloud service locally on Windows machines. The platform was recognized as one of **TIME's Best Inventions of 2021** for providing continuous education to children in conflict zones and refugee settings.
- **Global Societal Impact:** This "offline-first" prototype was the key technical catalyst that prompted **UNICEF** to launch the initiative using Microsoft Community Training for no-connectivity regions, contributing to a high-scale initiative that reached **7.4 million learners across 43 countries** by 2024;
- **Strategic Architectural Leadership:** Successfully advocated for, and worked on containerization and migration to **cross-platform .NET Core**; this move was critical for the platform's ability to run on diverse hardware in emerging markets like Zimbabwe, Lebanon, and Honduras.
- **Operational Support & Edge Strategy:** Facilitated the integration of **Azure IoT Hub** to support remote management and observability for offline learning hubs; assisted in testing data-sync on diverse Linux-based hardware to ensure reliable curriculum delivery in remote field operations.

Engineering Lead | Swachh Bharat Mission (powered by Project Sangam)

- Directed an engineering team through the end-to-end implementation of key mobile features, culminating in the successful launch of the official Swachh Bharat Mission eLearning mobile app.
- Spearheaded the rapid triage of launch-critical issues in close collaboration with Product Management, ensuring a smooth migration of the training platform.
- Architected and implemented a high-impact search-enabled dropdown to resolve a critical UI bug; this allowed users to efficiently navigate and select from **the hundreds of cities where the platform was deployed**.
- **Contributed to a high-scale social impact project** that successfully trained and empowered over **110,000+ municipal functionaries** across **4,000+ cities** in India.

Engineering Lead | Platform Localization

- Spearheaded global accessibility by **engineering an internal localization framework** and integrating with manual translation pipelines.
- Initially developed comprehensive support for **right-to-left (RTL) formats** before assuming full platform localization ownership.
- Conducted a **comprehensive investigation on ligature support** to ensure accurate rendering of complex scripts across the ecosystem.

Team Leadership & Mentorship: Mentored interns, full-time colleagues, and onboarding buddies; conducted technical design reviews and proactively collaborated with partner teams to align cross-functional engineering goals.

Designed a robust global load-balancing solution and BCDR solution using Azure Front Door, significantly improving page load performance and security.

High-Throughput Bulk Ingestion: Architected and built an end-to-end bulk upload system enabling parallel ingestion of multi-format media; significantly reduced administrative onboarding time and **super-charged the manual testing experience** for everyone.

Developed a scalable monitoring infrastructure using Event Hubs and Azure Data Explorer to centralize telemetry and enable diagnostic and analytic dashboards.

Automated end-to-end deployments and CI/CD pipelines, enabling the Public Preview release and publishing of the final offering on the Azure Marketplace.

Identity & Authentication: Streamlined the SMS-based login flow by implementing custom Identity Server solutions over Azure Cosmos DB and closely collaborated with Telesign to significantly reduce SMS code-sending costs across many countries.

Search Infrastructure & Optimization: Designed and implemented a custom search index featuring advanced schema optimizations and Lucene-based retrieval.

- Integrated with custom skills to extract metadata from PDFs, videos, and documents into indexed columns.
- Enhanced discovery by implementing **fuzzy matching, synonym mapping, and phonetic analyzers** using Azure Cognitive Search to improve query relevance.

Media Security & DRM: Authored a technical proposal to modernize content protection for temporal media using industry-standard **DRM (Digital Rights Management)** including **Widevine, PlayReady, and FairPlay** over existing/legacy AES-128 encryption.

End-to-End Notifications Infrastructure: Initially onboarded via Xamarin consumption APIs before assuming full ownership of the platform's notification subsystem, overseeing both backend delivery and frontend integration.

Advanced Thread Modeling: Conducted deep-dive threat modeling and attack surface analysis using Microsoft internal security frameworks; ensuring the integrity of user data and platform content.

Privacy & Compliance: Enabled automated compliance tooling to automate the identification of secure, licensed dependencies; ensuring rigorous adherence to global data protection and privacy standards.

Led the migration of 160M+ customer records from AWS to Azure, designing and implementing the necessary APIs to ensure a seamless transition.

State Persistence & User Analytics: Developed features for real-time tracking of user progress, including "most recently viewed lesson" and "last watched timestamp" for temporal media content ensuring a seamless cross-device learning experience.

Delivered end-to-end features and bug fixes spanning data layers, backend services, and frontend components.

The Garage, Microsoft

May 2016 - July 2016

Software Engineering Internship — Hyderabad, Telangana, India

- Engineered and deployed Azure-hosted chat-bots using the **Microsoft Bot Framework**, specifically architecting custom conversational flows for automated form-filling. Helped with OpenCV usage for improved OCR performance and test Microsoft's LUIS for a more natural language experience.

Xerox Research Centre, India

Dec 2015 - Jan 2016

Machine Learning Internship — Bengaluru, Karnataka, India

- **Xerox Research Innovation Challenge:** Engineered predictive neural network models using vitals and lab data to forecast patient mortality and health complications among ICU patients.
- Secured **3rd place** (online) and **7th place** (offline) in the challenge alongside a 3-person team, successfully competing against advanced Ph.D. research teams.
- Completed an intensive Winter School on Machine Learning under the mentorship of Prof. Vaibhav Rajan.

Projects

- **AIMA-CSharp Open Source Contribution:** Contributed to the AIMA-CSharp repository by implementing pseudo-code in C# from Peter Norvig's *Artificial Intelligence: A Modern Approach*.
- **Image Steganography:** Implemented a system to encrypt and embed hidden messages within images.
- **IoT Home Automation:**: Team project to retrofit legacy electric switches with a motorized 3D-printed actuators and control software.
- **Kool AI(d) for Gamers:** Extracted live game metrics and game states from DotA-2 streams for real-time analytics.
- **The Chess Game:** A chess game with AI using Mini-Max and Alpha-Beta pruning, with multiplayer support.
- **Multi-Agent Environment Simulation:** Built a 2D grid simulation of agents coordinating with one another.
- **2D Physics Game Engine:** Developed a custom rigid-body and gravitational physics engine in Java focusing on real-time kinematic interactions.
  - Implemented efficient **collision detection and resolution** for multiple dynamic objects.
  - Engineered an **environmental boundary system** to manage object interactions with surrounding walls.
  - Implemented a **Near-Earth constant gravity model** to simulate realistic downward acceleration.
  - Implemented a **Newtonian** $1/r^2$ **gravity model** to simulate object interaction in space.
- **Intel E5200 CPU Overclocking**: Successfully overclocked an Intel Pentium E5200 processor from a base clock of **2.5GHz to 2.8GHz** to improve gaming performance on budget hardware.

- Optimized system performance by manually adjusting **FSB frequencies, CPU multipliers, and core voltage settings** within the BIOS to achieve a stable 12% frequency increase.
- Conducted extensive stress testing to monitor thermal output and system stability under peak loads, ensuring long-term hardware reliability and optimal heat dissipation.
- Benchmarked and Documented performance gains on hardware-intensive titles such as Grand Theft Auto IV.

- **Top-Down Space Shooter (Unity):** Developed a 2D top-down space shooter game using the Unity game engine.
- **Predicting Tourism Trends:** Worked on building statistical and machine learning models to forecast tourism trends.
- **Robo-Soccer:** Placed runner-up in a team of 5 at a Robo-Soccer event organized by Technex, IIT (BHU) Varanasi.
- **Tic-Tac-Toe AI:** Implemented a decision-tree based AI and multi-player for Tic-Tac-Toe.

Education

Indian Institute of Technology (BHU) - Varanasi
July 2013 - May 2018

Integrated Master's degree (5 years): Mathematics & Computing — Varanasi, India

- **Problem Setter & Moderator – Hackerrank (Technex'16):**
  - **Mathletics:** www.hackerrank.com/mathletics-technex16
  - **Prayaas:** www.hackerrank.com/prayaas-technex16
- **Teaching Assistant** for the freshman course on **Mathematical Methods**.
- Thesis: Deep Learning Based Classification of Handwritten Mathematical Symbols (Supervised by Prof. T. Som)
- Skills/Technologies used: Pandas, NumPy, TensorFlow, Keras, CNNs, Scikit-learn, AWS etc.
- Trained transfer learning models achieving state-of-the-art accuracy on the HasyV2 dataset.

- **DeepLearning.AI:** Deep Learning Specialization — *August 2017*
- **University of Alberta & Amii:** Reinforcement Learning Specialization — *July 2020*
- **The State University of New York:** Blockchain Basics — *June 2021*
- **DeepLearning.AI:** Generative AI For Software Development — *February 2025*

Accomplishments

- **Competitive Programming, Machine Learning and Quant**
  - Qualified for and represented IIT Varanasi at **ACM-ICPC 2016 Regionals**
  - Qualified for and represented IIT Varanasi at **ACM-ICPC 2015 Regionals**
  - Ranked 15th worldwide (2nd in India) in **Codeforces Round 410**
  - Finalist of GS Quantify 2016 (2nd in India), a flagship event, organized by **Goldman Sachs**
  - Ranked 1st in COPS-OPC-2015, a programming contest organized by the club of programmers at IIT (BHU)
  - Ranked 1st in CodeDef 2016, a coding contest organized in Udyam, a college festival of IIT (BHU)
  - Ranked 14th out of 3,000+ participants in the IndiaHacks Machine Learning track
  - Ranked 2nd in Analiticity, a campus-wide Data Science competition at Technex '16
  - Ranked 83rd worldwide in **Google Kickstart** Round D 2017

- **Academic**
  - All India Rank 1672 in IIT-JEE (ADVANCED) - 2013, among 150k candidates
  - All India Rank 494 in Joint Entrance Examination - JEE (MAINS) - 2013, among 1.3 million candidates
  - All India Rank 6355 in National Eligibility cum Entrance Tests (NEET) - 2013, among 650k candidates
  - Achieved Rank 1 in Medical stream of Jharkhand Combined Entrance Competitive Examination - 2013
  - Achieved Rank 2 in Engineering stream of Jharkhand Combined Entrance Competitive Examination - 2013
  - Awarded DST INSPIRE Science Scholarship for ranking in the top 1% of AISSCE 2013
  - Scored 96.8% in the All India Senior School Certificate Examination
  - Recognized in the top 0.1% of AISSCE 2013 for excellence in Physical Education

SKILLS

- **Programming Languages:** C, C++, C#, Java, Python, TypeScript, Scala, Hack/PHP, SQL, PowerShell, Octave
- **Backend & Distributed Systems:** High-Scale Services (Billions of daily requests), Distributed Systems Architecture, API Design, Data Pipelines, Authentication & Identity Systems, Blockchain, BCDR (Business Continuity & Disaster Recovery).
- **Cloud & Infrastructure:** Microsoft Azure (Expert), AWS, Kubernetes, CI/CD & Infrastructure Automation.
- **Data & Storage:** Blockchain, NoSQL/SQL, Redis, Search Infrastructure (Lucene/Elasticsearch).
- **Frameworks & Libraries:** .NET / ASP.NET Core, React, TensorFlow, Keras, Pandas, NumPy, Scikit-learn, XGBoost, OpenCV, Unity, IronPDF.
- **AI / Machine Learning:** LLM Applications & Agentic Workflows, Reinforcement Learning, Deep Learning (CNNs), Transfer Learning, Neural Networks, Computer Vision.
- **Engineering Leadership:** Technical Design Reviews, Mentorship & Team Leading, On-call Engineering & Incident Response, Security & Privacy Reviews, Threat Modeling, Cross-functional Collaboration.
- **Natural Languages:** English and Hindi.